

Title: REG4 marks a lineage-restricted secretory epithelial state in gastrointestinal adenocarcinomas: a pan-cancer state-dominance analysis

Abstract

Background: Regenerating islet-derived protein 4 (REG4) is overexpressed in several gastrointestinal (GI) cancers and has been linked to immune infiltration, stromal remodelling and patient outcome. However, these associations have mostly been studied one cancer type at a time, without accounting for epithelial cell state or tumour purity. We asked which biological axis REG4 primarily tracks across human cancers.

Methods: We analysed batch-corrected RNA-seq data from 9,849 primary tumours in 34 TCGA PanCanAtlas cohorts. Within each cohort, REG4 expression was correlated with epithelial/secretory, stromal/EMT, immune/cytotoxic and natural killer (NK) cell gene-expression scores, with and without adjustment for tumour purity, and a dominant axis was assigned using pre-specified criteria. Histology, molecular subtype and survival were examined in cohorts where REG4 was secretory-associated.

Results: The epithelial/secretory axis was dominant in 5 of 6 GI adenocarcinoma cohorts (ρ 0.62–0.76) compared with 2 of 28 other cohorts (odds ratio 49, $p = 4.3 \times 10^{-4}$), and this enrichment persisted after purity adjustment ($p = 0.003$). REG4 was highest in mucinous colorectal and gastric tumours and tracked the secretory program even among non-mucinous GI tumours (ρ 0.63–0.88). In contrast, mucinous breast carcinomas activated the secretory program without expressing REG4. The apparent association between REG4 and immune infiltration in colorectal cancer was largely explained by microsatellite-instability (MSI) subtype, and REG4 added no prognostic information beyond secretory state (pooled GI hazard ratio 0.96 per SD, $p = 0.56$).

Conclusions: REG4 is not a pan-cancer immune or stromal marker. It marks a lineage-restricted secretory epithelial state in GI adenocarcinomas, and its immune and prognostic associations reflect tumour subtype and cell state rather than an independent REG4-linked program.

Keywords: REG4; gastrointestinal adenocarcinoma; mucinous; secretory lineage; pan-cancer; TCGA; microsatellite instability

Regenerating islet-derived protein 4 (REG4) is a secreted C-type lectin and the most recently identified member of the regenerating (REG) gene family. In the normal gastrointestinal (GI) tract, REG4 is restricted to secretory epithelial lineages. In the colon, it marks deep crypt secretory cells, which act as Paneth-cell equivalents that support Lgr5⁺ stem cells. In the stomach, it is expressed in goblet cells of intestinal metaplasia, and it is a direct transcriptional target of the intestinal master regulator CDX2. REG4 is overexpressed in several GI malignancies, including colorectal, gastric and pancreatic adenocarcinomas, and is a useful immunohistochemical marker of signet-ring cell carcinoma. Outside the GI tract, REG4 has been reported mainly in prostate cancer, where it is enriched in hormone-refractory metastatic disease.

Beyond its epithelial expression, REG4 has been linked to a broad and sometimes inconsistent set of tumour features, including patient outcome, chemotherapy response, immune infiltration and stromal remodelling. Most of these associations were derived from single cancer types, and many from bulk transcriptomic correlations. Such correlations are vulnerable to two well-recognised confounders. First, the expression of any epithelium-restricted gene co-varies with tumour purity, so that it correlates positively with epithelial programs and negatively with immune and stromal content regardless of its function. Second, molecular subtypes bundle several phenotypes together. For example, microsatellite-unstable (MSI) colorectal cancers are frequently mucinous and also densely infiltrated by lymphocytes, so a gene tracking mucinous differentiation could appear "immune-associated" without having any immunological role. Whether the reported immune, stromal and

prognostic associations of REG4 reflect REG4-specific biology, or simply the cellular state in which REG4 is expressed, has not been tested.

A related gap is lineage. If REG4 primarily marks a GI secretory state, its associations should be coherent and strong in GI adenocarcinomas but weak or inconsistent elsewhere, including in non-GI tumours that activate mucin programs. This prediction requires a common analytical framework applied across many cancer types. Specifically, one must ask which biological axis REG4 tracks most strongly in each tumour type, whether that axis survives adjustment for purity, and whether REG4 carries information beyond the state it marks. This distinction matters. A passive marker of secretory state is useful for classification, whereas a gene that independently predicts immune features or outcome within that state would merit functional and clinical investigation.

Here, we hypothesised that REG4's associations with immune and stromal features vary across cancers, but that its most coherent biology is epithelial/secretory state identity, particularly in GI-lineage adenocarcinomas. To test this, we analysed 9,849 primary tumours from 34 TCGA PanCanAtlas cohorts. Within each cohort, we correlated REG4 with epithelial/secretory, stromal/EMT, immune/cytotoxic and natural killer (NK) cell programs, with and without adjustment for ABSOLUTE tumour purity, and assigned a dominant REG4-associated axis using pre-specified criteria. We then examined mucinous histology (including a non-GI mucinous control), molecular subtype and survival, to distinguish REG4 as a marker of secretory state from REG4 as an independent correlate of immune features or outcome.

Methods

Data sources

Batch-corrected RNA-seq gene expression data (EB++AdjustPANCAN_IlluminaHiSeq_RNASeqV2) from the TCGA PanCanAtlas were obtained from the UCSC Xena PanCanAtlas hub using the UCSCXenaTools R package. The data are RSEM-normalised, batch-corrected across sequencing centres and cancer types, and log2-transformed. Only the genes required for the analysis were retrieved. Clinical and survival data, including cancer type, age at diagnosis, AJCC pathological stage, histological type, overall survival (OS) and progression-free interval (PFI), were taken from the curated TCGA Pan-Cancer Clinical Data Resource (TCGA-CDR). Tumour purity estimates were obtained from the PanCanAtlas ABSOLUTE analysis. Molecular subtypes for GI adenocarcinomas (chromosomal instability, CIN; genomically stable, GS; hypermutated-indel/microsatellite-unstable, MSI; hypermutated-SNV/POLE; and Epstein–Barr virus-positive, EBV) were obtained from the PanCanAtlas subtype annotation (TCGASubtype.20170308, "Subtype_Selected").

Sample selection and cohort definition

Only primary tumour samples were included (TCGA sample type 01, or 03 for acute myeloid leukaemia), and one sample was retained per patient. After merging with clinical data, the final dataset comprised 9,849 primary tumours. Oesophageal carcinomas (ESCA) were separated by histology into adenocarcinoma (ESCA_EAC, n = 89) and squamous cell carcinoma (ESCA_ESCC, n = 95), giving 34 cohorts. ABSOLUTE purity estimates were available for 95.3% of tumours.

Cohorts were assigned to three groups:

- **GI adenocarcinomas** (n = 6): colon (COAD), rectal (READ), pancreatic (PAAD), stomach (STAD), oesophageal adenocarcinoma (ESCA_EAC) and cholangiocarcinoma (CHOL).
- **Non-epithelial tumours** (n = 12): LAML, DLBC, THYM, SARC, SKCM, UVM, GBM, LGG, PCPG, TGCT, MESO and UCS.
- **Other epithelial tumours** (n = 16): all remaining cohorts.

Gene sets and program scores

Six gene sets were defined *a priori*:

- **Secretory:** MUC2, TFF3, SPINK4, FCGBP, AGR2
- **Epithelial:** KRT8, KRT18, KRT19, EPCAM
- **GI lineage transcription factors:** CDX2, VIL1, KLF4, HNF4A
- **Stromal/EMT:** COL1A1, COL1A2, COL3A1, COL5A1, FN1, SPARC, POSTN, FAP, PDGFRB, THY1, DCN, LUM, VIM, ZEB1, SNAI2, TWIST1
- **Immune/cytotoxic:** PTPRC, CD2, CD3D, CD3E, CD8A, LCK, GZMA, GZMB, PRF1, IFNG, CXCL9, CXCL10
- **Natural killer (NK):** KLRD1, KLRF1, NCR1, NCR3, SH2D1B, KIR2DL4, KIR3DL1, NCAM1

The epithelial/secretory score combined the secretory and epithelial sets (nine genes). REG4 was excluded from all scores. For each cohort separately, the expression of each gene was converted to a z-score, and each program score was calculated as the mean z-score of its member genes. Because scores were standardised within cohorts, all correlations compare tumours within the same cancer type rather than across cancer types. Genes with missing values in a cohort were omitted from the affected scores.

REG4 detection filter

REG4 was considered detected in a tumour when its log₂ expression exceeded 1. Cohorts in which REG4 was detected in fewer than 20% of tumours (10 cohorts) were classified as "not expressed" and excluded from dominance calls and group comparisons, because correlations in these cohorts are driven by a small minority of expressing tumours.

Correlation analyses

Within each cohort, the association between REG4 and each marker gene or program score was assessed by Spearman rank correlation. To control for tumour purity, partial Spearman correlations adjusted for ABSOLUTE purity were calculated with the ppcor package. Because the immune/cytotoxic and NK scores were strongly correlated ($\rho = 0.76$), the association of REG4 with NK-specific signal was tested separately as a partial Spearman correlation between REG4 and the NK score, adjusted for the immune/cytotoxic score. P values were corrected for multiple testing using the Benjamini–Hochberg false discovery rate (FDR), applied across all tests for gene-level analyses and within each axis for program-score analyses.

Dominant-axis assignment

For each cohort, the dominant REG4-associated axis was selected from three primary axes: epithelial/secretory, stromal/EMT and immune/cytotoxic. The axis with the largest absolute correlation was called dominant if all three of the following criteria were met: $|\rho| \geq 0.3$, $\text{FDR} < 0.05$, and an absolute correlation at least 0.1 greater than that of the second-ranked axis. Otherwise, the cohort was classified as having no dominant axis. Dominance was assigned three ways: using raw correlations (primary analysis), using purity-adjusted partial correlations, and replacing the epithelial/secretory score with the secretory-only score (sensitivity analyses). A call was considered purity-robust when the raw and purity-adjusted analyses assigned the same axis.

Comparison of GI and non-GI tumours

Enrichment of epithelial/secretory dominance in GI adenocarcinomas was tested with Fisher's exact test in three ways: across all 34 cohorts, across REG4-expressing cohorts only, and using purity-adjusted dominance calls. Among expressing cohorts, the correlations between REG4 and the epithelial/secretory score were compared between GI and non-GI cohorts using the Wilcoxon rank-sum test.

Histological analysis

Tumours were classified as mucinous when their TCGA histological type included the terms "mucinous", "signet ring" or "colloid". Within each cohort with at least three mucinous tumours, REG4 expression and secretory score were compared between mucinous and non-mucinous tumours using the Wilcoxon rank-sum test. To determine whether the association between REG4 and the secretory program depended on histology, Spearman correlations between REG4 and the secretory score were recalculated after excluding all mucinous tumours.

Survival analysis

Survival analyses were restricted to cohorts in which REG4 was epithelial/secretory-dominant (COAD, READ, PAAD, STAD, ESCA_EAC, BLCA and PRAD). PFI was the primary endpoint and OS the secondary endpoint. REG4 expression was standardised within each cohort, so hazard ratios (HR) are expressed per standard deviation (SD). For each cohort and endpoint, a base Cox proportional hazards model included the epithelial/secretory score and age, plus stage (I–II versus III–IV) where stage was available for at least 70% of patients. A full model added REG4. The additional prognostic contribution of REG4 was assessed with a likelihood-ratio test comparing the two models, and discrimination was summarised with Harrell's concordance index (C-index). Models were fitted only when at least 50 patients and 15 events were available. Likelihood-ratio P values were corrected using the Benjamini–Hochberg method. A pooled model across the five GI adenocarcinoma cohorts was fitted for PFI, stratified by cohort and adjusted for the epithelial/secretory score and age.

Four-state analysis

Within each cohort, tumours were classified as REG4-high or REG4-low, and as secretory-high or secretory-low, using the cohort median of each variable. Combining the two splits gave four states: REG4-low/secretory-low, REG4-high/secretory-low, REG4-low/secretory-high and REG4-high/secretory-high. Median immune/cytotoxic, NK, stromal/EMT and purity values were compared across states. To test whether REG4 carried information beyond the secretory program, linear regression models were fitted for each of the three microenvironment scores, with standardised REG4, standardised secretory score, their interaction and tumour purity as predictors. P values were corrected using the Benjamini–Hochberg method. PFI was compared across the four states using Cox models adjusted for age, with REG4-low/secretory-low as the reference, in cohorts with at least 15 events and at least five tumours per state.

Molecular subtype adjustment

To determine whether the association between REG4 and immune features in GI tumours reflected molecular subtype, linear models of the immune/cytotoxic and NK scores were fitted in COAD, READ and STAD with standardised REG4, standardised secretory score and purity as predictors, with and without molecular subtype as an additional covariate. These analyses were restricted to tumours with subtype annotation.

Statistical analysis and software

All analyses were performed in R (version 4.5.1) using the `data.table`, `survival`, `ppcor`, `ggplot2`, `patchwork` and `UCSCXenaTools` packages. All tests were two-sided, and $FDR < 0.05$ was considered significant unless otherwise stated.

Code availability

The complete analysis pipeline, which reproduces all results and figures from the public data sources, is available at [GitHub repository URL].

Results

REG4 expression is lineage-restricted and highest in GI adenocarcinomas

We first examined REG4 expression across 9,849 primary tumours from 34 cohorts (Figure 1A). Five GI adenocarcinoma cohorts (ESCA_EAC, STAD, COAD, PAAD and READ) expressed REG4 at uniformly high levels (median log₂ expression 9.91–11.12), with REG4 detected in 97–98% of tumours. PRAD was the only non-GI cohort with comparable detection (99%), but its expression level was lower (median 5.74), approximately 18- to 42-fold below the GI cohorts. In all remaining cohorts, median REG4 expression was 2.21 or lower, at least 200-fold below the GI adenocarcinomas. Cholangiocarcinoma (CHOL) was the exception within the GI group (median 1.23; detected in 61% of tumours). REG4 was essentially absent, detected in fewer than 20% of tumours, in 10 cohorts (ACC, UVM, THCA, SKCM, MESO, PCPG, GBM, SARC, KIRP and KIRC); these were excluded from downstream correlation analyses. In several REG4-low epithelial cohorts, most notably LUAD and CESC, a small minority of tumours reached expression levels comparable to GI adenocarcinomas, possibly reflecting rare tumours with GI-type differentiation. REG4 expression is therefore strongly lineage-restricted. This motivated a within-cohort analysis of which biological program REG4 tracks in the tumours where it is expressed.

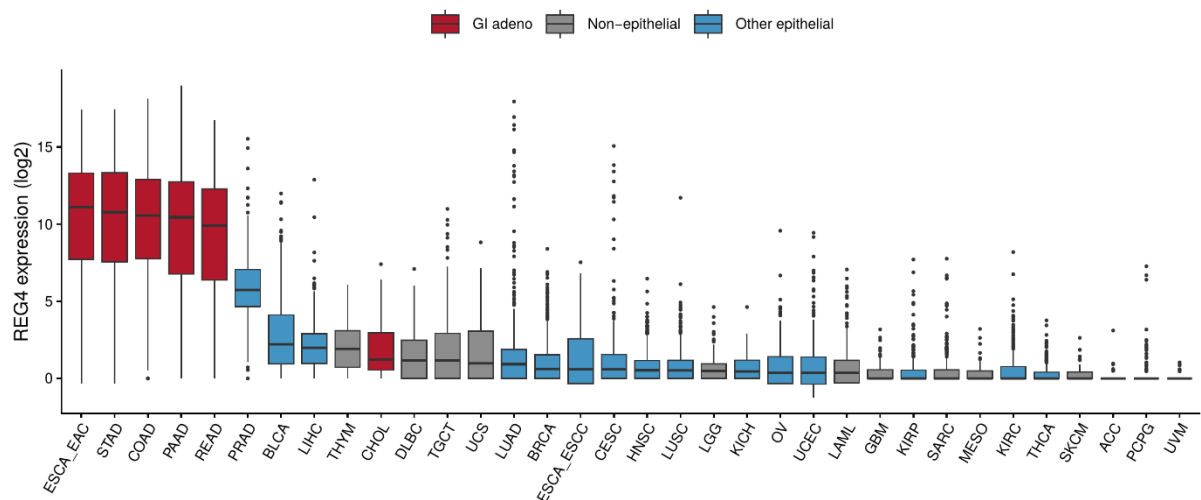


Figure 1A. REG4 expression across 34 TCGA cohorts. Boxplots show REG4 expression (log₂, batch-corrected) in 9,849 primary tumours, ordered by cohort median. Colours indicate cohort group: GI adenocarcinoma (red), other epithelial (blue) and non-epithelial (grey). Boxes show the median and interquartile range; whiskers extend to 1.5 times the interquartile range; points indicate outlying tumours. Cohort sample sizes and detection rates are given in Supplementary Table S1.

REG4 is dominantly associated with the epithelial/secretory axis in GI adenocarcinomas

To determine which biological program REG4 primarily tracks, we correlated REG4 with epithelial/secretory, stromal/EMT, immune/cytotoxic and NK scores within each cohort and assigned a dominant axis using pre-specified criteria (Figure 1B). The four scores behaved as largely independent axes, except for the immune/cytotoxic and NK scores, which were strongly correlated with each other ($\rho = 0.76$) and were therefore handled separately (see Methods).

The epithelial/secretory axis was dominant in five of the six GI adenocarcinoma cohorts: READ ($\rho = 0.76$), COAD (0.72), PAAD (0.68), ESCA_EAC (0.64) and STAD (0.62). In each of these cohorts, the epithelial/secretory correlation exceeded the next-strongest axis by a wide margin (0.38–0.61), and all five calls were retained both after adjustment for tumour purity (partial ρ 0.61–0.76) and when the secretory-only score was used. These were the only dominance calls in the dataset that were unchanged across all three versions of the analysis. CHOL showed no dominant axis (epithelial/secretory $\rho = 0.14$, FDR = 0.49), consistent with its low REG4 expression and small sample size ($n = 36$).

Outside the GI tract, REG4 associations were weaker and less consistent. Among the 18 non-GI cohorts that expressed REG4, only two were epithelial/secretory-dominant: BLCA ($\rho = 0.49$) and PRAD ($\rho = 0.32$). Both calls were purity-robust, although the BLCA call was lost when the secretory-only score was used. Four cohorts were immune-dominant at modest strength (ρ 0.30–0.34): DLBC and three squamous carcinomas, HNSC, ESCA_ESCC and LUSC. Only the LUSC call survived purity adjustment, suggesting that the low-level REG4 signal in squamous tumours partly reflects tumour purity rather than a tumour-intrinsic program. OV was classified as stromal-dominant on the basis of a negative correlation ($\rho = -0.31$), which was lost after purity adjustment. TGCT showed comparable epithelial/secretory ($\rho = 0.52$) and stromal ($\rho = 0.50$) correlations and therefore had no dominant axis. The remaining expressing cohorts showed no dominant axis, and the 10 non-expressing cohorts were not classified.

Epithelial/secretory dominance was strongly enriched in GI adenocarcinomas (5/6 GI cohorts versus 2/28 other cohorts; Fisher's exact test, odds ratio [OR] = 49.2, $p = 4.3 \times 10^{-4}$). The enrichment remained significant when the analysis was restricted to REG4-expressing cohorts (5/6 versus 2/18; OR = 30.4, $p = 0.0027$) and when purity-adjusted dominance calls were used (OR = 30.4, $p = 0.0027$). Consistent with this, the median correlation between REG4 and the epithelial/secretory score was 0.66 in GI adenocarcinomas, compared with 0.17 in other epithelial and 0.16 in non-epithelial cohorts (Wilcoxon rank-sum test, $p = 0.0044$). Raw and purity-adjusted correlations were closely concordant across all expressing cohorts, and the GI adenocarcinomas lay on the line of identity (Figure S2), confirming that the epithelial/secretory association was not driven by tumour purity.

Together, these results show that REG4 is not a pan-cancer immune or stromal marker. Its dominant association is with the epithelial/secretory program, and this association is specific to GI adenocarcinomas.

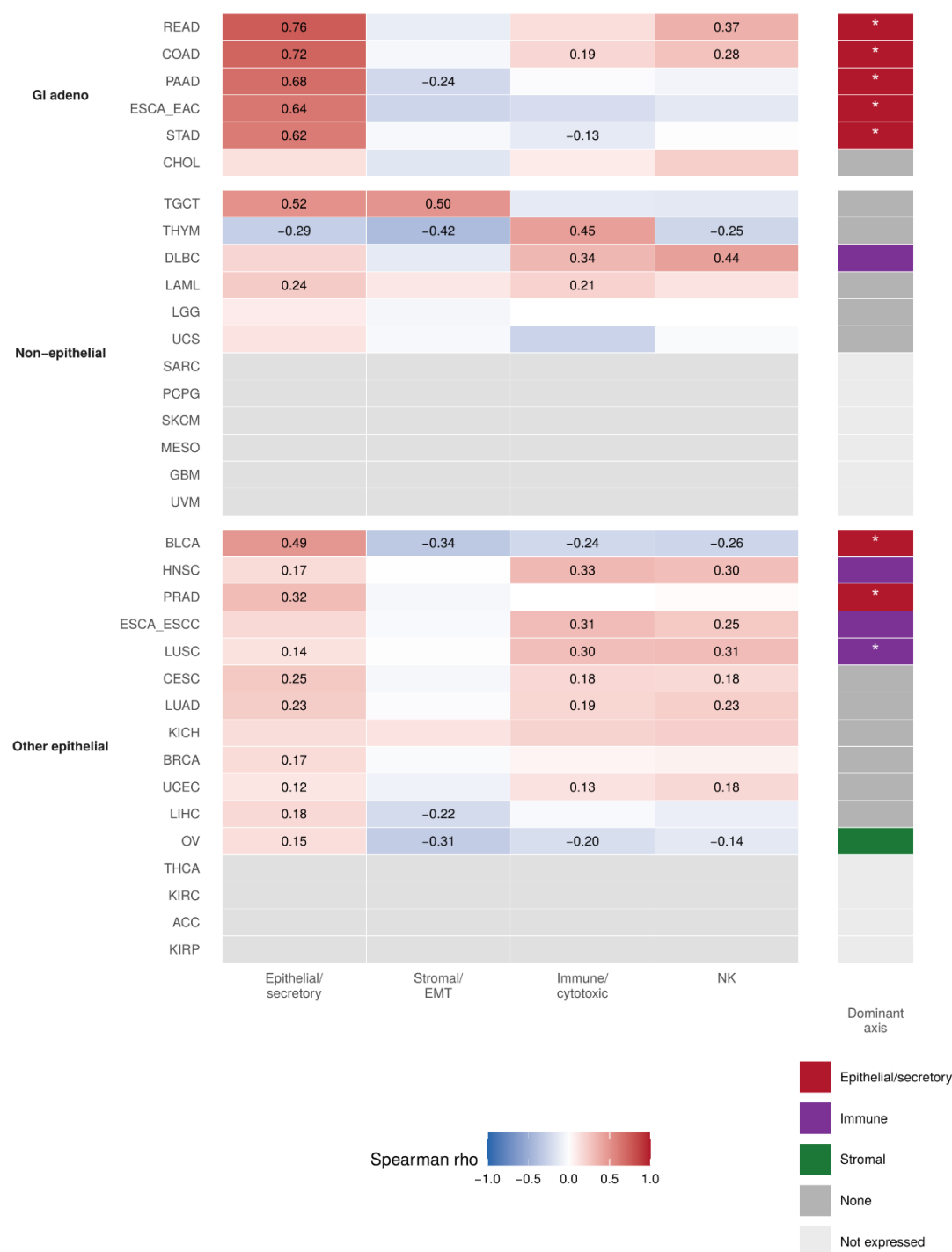


Figure 1B. Dominant REG4-associated axis across 34 TCGA cohorts. The heatmap shows the Spearman correlation between REG4 and four program scores (epithelial/secretory, stromal/EMT, immune/cytotoxic and NK) within each cohort. Values are shown for correlations with FDR < 0.05. The right-hand column shows the dominant axis assigned to each cohort ($|\rho| \geq 0.3$, FDR < 0.05 and a margin of at least 0.1 over the second-ranked axis). An asterisk (*) indicates that the dominance call was retained after adjustment for ABSOLUTE tumour purity. Grey rows indicate cohorts in which REG4 was detected in fewer than 20% of tumours ("not expressed").

S2

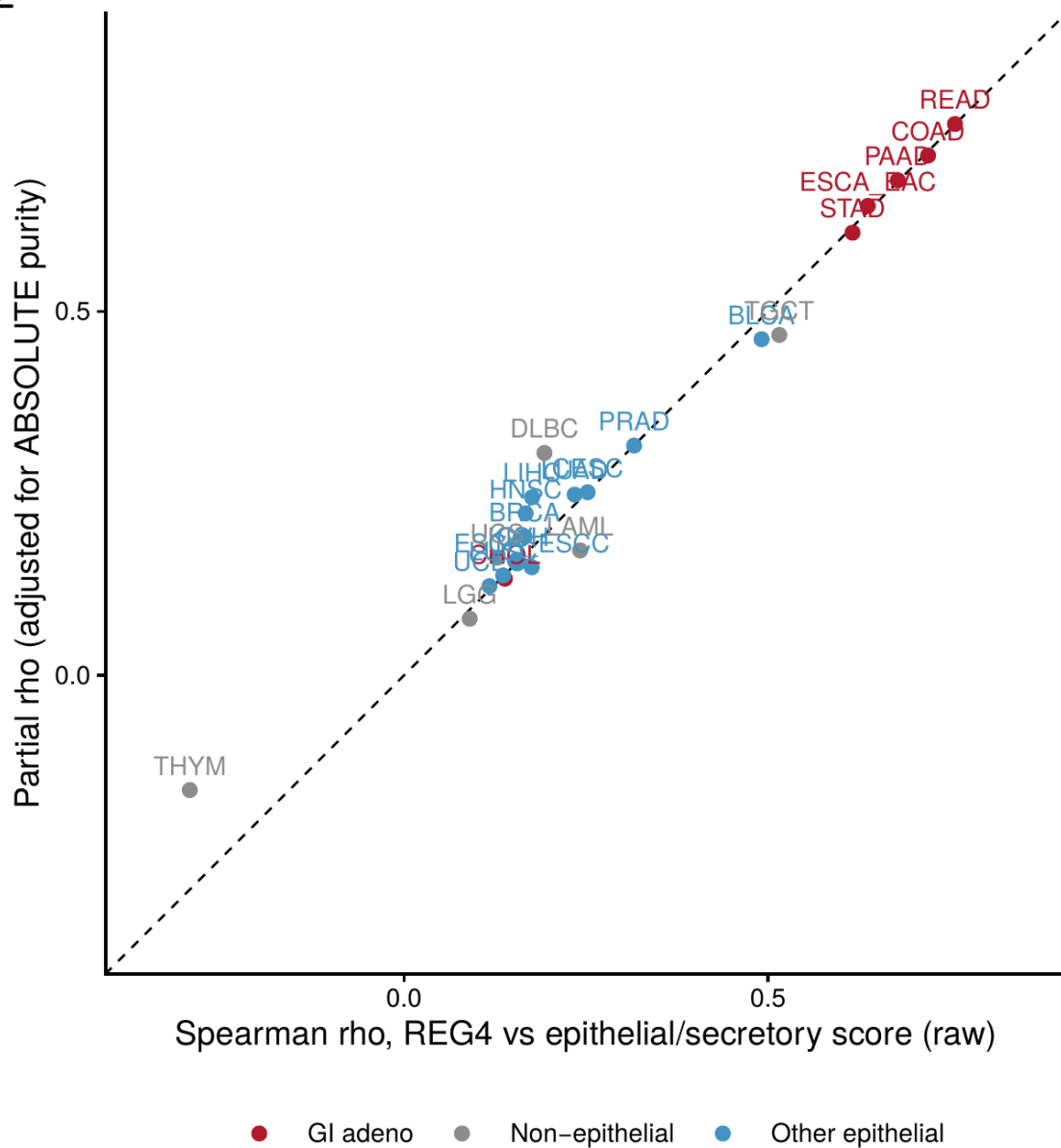


Figure S2. Purity robustness of REG4–epithelial/secretory correlations. Raw Spearman correlations between REG4 and the epithelial/secretory score (x-axis) are plotted against partial correlations adjusted for ABSOLUTE tumour purity (y-axis) for the 24 REG4-expressing cohorts. The dashed line indicates identity. Colours indicate cohort group.

REG4 tracks the goblet/secretory program, and its coupling to intestinal lineage factors depends on the organ

To define which components of the epithelial/secretory program drive the REG4 association, we examined REG4 correlations with the individual secretory, epithelial and GI lineage transcription factor genes (Figure 2A).

In all five REG4-high GI adenocarcinomas, REG4 correlated most strongly with genes of the goblet/secretory program. The strongest associations were in the colorectal cohorts, where REG4 correlated with SPINK4 ($\rho = 0.90$ in READ and 0.79 in COAD), MUC2 (0.87 and 0.74), FCGBP (0.84 and 0.68) and AGR2 (0.68 and 0.75). Correlations with these genes were weaker but consistently positive in STAD (0.36 – 0.57), PAAD (0.33 – 0.62) and ESCA_EAC (0.45 – 0.64). TFF3 showed moderate correlations across GI cohorts (0.33 – 0.54), probably because it is broadly expressed in GI tumours. At the level of the combined secretory score, correlations ranged from 0.64 to 0.88 across the four largest GI cohorts (Figure 2B).

In contrast, correlations with general epithelial genes were weak. Across the GI cohorts, REG4 correlations with KRT8, KRT18 and KRT19 were mostly between 0.1 and 0.3 , and REG4 was negatively correlated with EPCAM in the colorectal cohorts ($\rho = -0.24$ in COAD and -0.19 in READ). REG4 therefore does not simply track epithelial content. Instead, it marks a specific secretory state within the epithelial compartment.

The relationship between REG4 and the GI lineage transcription factors differed sharply by organ. In the colorectal cohorts, REG4 was negatively correlated with the intestinal identity factors CDX2 ($\rho = -0.41$ in COAD and -0.34 in READ), HNF4A (-0.41 and -0.34) and VIL1 (-0.35 and -0.26 ; all $\text{FDR} < 0.005$). At the same time, it was positively correlated with KLF4, a transcription factor required for goblet-cell differentiation (0.59 and 0.62). In PAAD, the pattern was reversed: REG4 was positively correlated with CDX2 (0.52), VIL1 (0.57) and HNF4A (0.51 ; all $\text{FDR} < 10^{-11}$). STAD showed intermediate, weakly positive correlations (CDX2 0.28 , HNF4A 0.21). Thus, in the colon, where intestinal identity is native, high REG4 marks tumours shifted toward a goblet-like secretory state and away from the canonical enterocyte program. In the pancreas and stomach, where intestinal identity is not native, high REG4 marks tumours that have acquired intestinal-type differentiation. In CHOL, REG4 was not significantly correlated with the secretory genes but was correlated with CDX2 ($\rho = 0.49$, $\text{FDR} = 0.007$), suggesting that the few REG4-expressing cholangiocarcinomas may have intestinal-type features.

Outside the GI tract, REG4 was largely decoupled from this program. In PRAD, despite high REG4 expression, REG4 correlated modestly with MUC2 (0.40) and AGR2 (0.30) but not with SPINK4, TFF3 or any GI lineage factor (all $|\rho| \leq 0.11$), and its correlation with the secretory score was weak ($\rho = 0.34$; Figure 2B). In BRCA, REG4 showed essentially no relationship with the secretory score ($\rho = 0.17$). BLCA was the only non-GI cohort with moderate correlations across several secretory genes (0.35 – 0.43).

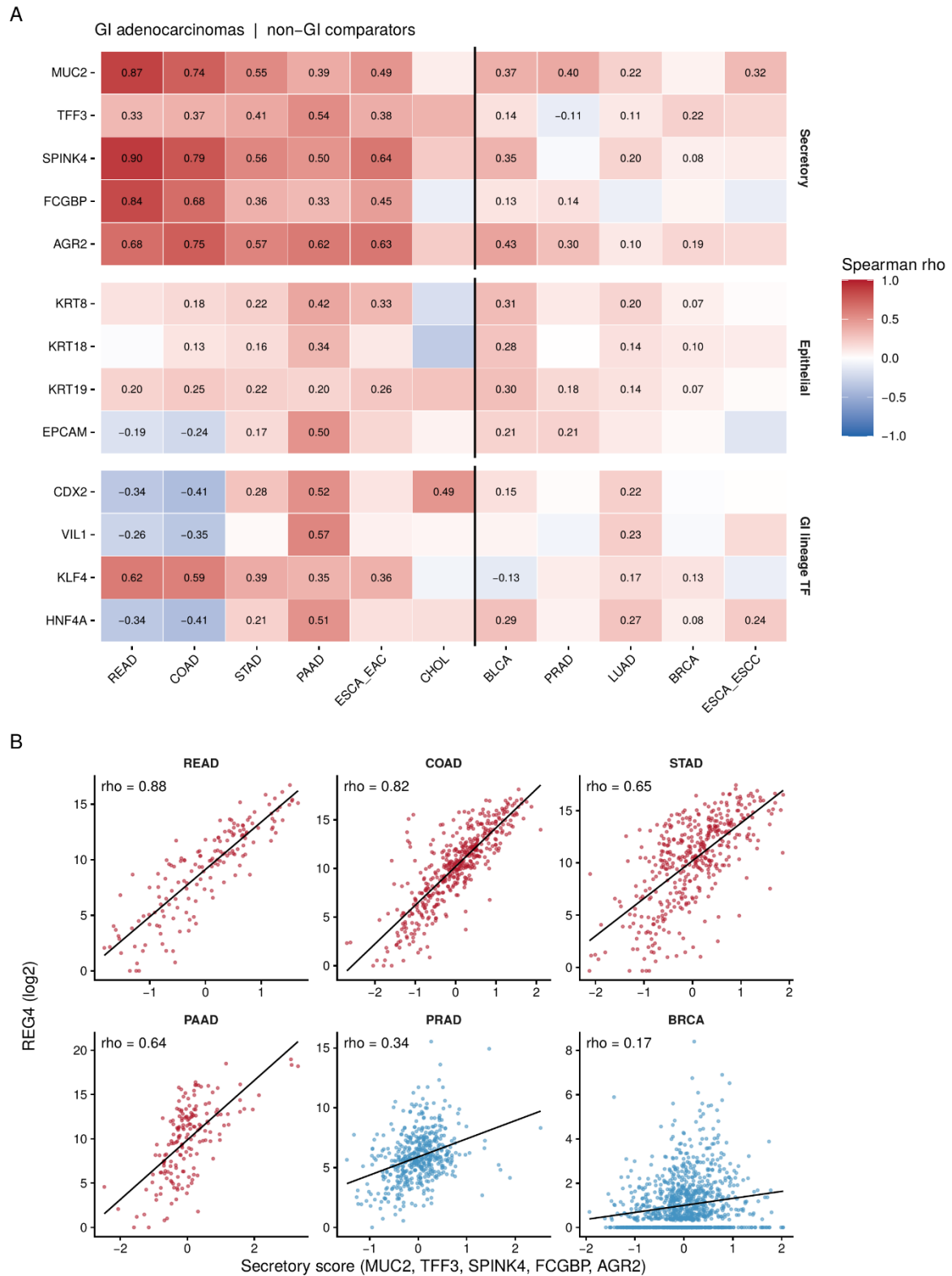


Figure 2. REG4 tracks the goblet/secretory program, with organ-dependent coupling to GI lineage factors. (A) Spearman correlations between REG4 and 13 marker genes (secretory, epithelial and GI lineage transcription factor sets) in GI adenocarcinoma cohorts (left of the vertical line) and non-GI comparator cohorts (right). Values are shown for correlations with FDR < 0.05. (B) REG4

expression plotted against the secretory score (mean z-score of MUC2, TFF3, SPINK4, FCGBP and AGR2) in four GI cohorts (red) and two non-GI cohorts (blue). Lines show linear fits; Spearman rho is shown in each panel.

REG4 marks a graded mucinous state that requires GI lineage context

We next asked whether REG4 tracks mucinous differentiation defined by histology, and whether this relationship holds outside the GI tract (Figure 3).

Within the GI cohorts, REG4 was consistently highest in mucinous tumours (Figure 3A). In COAD, mucinous adenocarcinomas ($n = 61$) expressed REG4 at a median log2 level of 14.29, compared with 10.27 in non-mucinous tumours (approximately 16-fold higher; $p = 2.4 \times 10^{-12}$). READ showed the same difference (13.57 versus 9.59 in 13 mucinous tumours; $p = 1.8 \times 10^{-5}$). In STAD, mucinous and signet-ring carcinomas ($n = 32$) expressed REG4 at 12.25, compared with 10.60 in other tumours (approximately 3-fold higher; $p = 0.004$); signet-ring (median 12.48) and mucinous-type (12.11) tumours had the highest levels of any gastric histology. In PAAD, the four colloid carcinomas expressed higher REG4 than ductal adenocarcinomas (13.05 versus 10.38), although this comparison was underpowered ($p = 0.11$). As expected, the secretory score was also elevated in mucinous tumours in every GI cohort (Figure 3B).

To determine whether REG4 simply reflects histological category, we repeated the correlation analysis after excluding all mucinous tumours (Figure 3C). REG4 remained strongly correlated with the secretory score among non-mucinous tumours in READ ($\rho = 0.88$), COAD (0.80), ESCA_EAC (0.68), STAD (0.64) and PAAD (0.63). REG4 therefore tracks a continuous, graded secretory state that extends into tumours classified by pathologists as conventional adenocarcinoma, rather than marking the mucinous histological category alone. In non-GI cohorts, the corresponding correlations remained weak (BLCA 0.43; OV, BRCA, LUAD and UCEC 0.04–0.23).

Mucinous breast carcinoma provided a critical negative control. These tumours ($n = 17$) strongly activated the secretory program, with a median secretory score of 1.38 compared with 0.00 in other breast cancers ($p = 9 \times 10^{-12}$), a level comparable to mucinous GI tumours. However, REG4 was not expressed in mucinous breast carcinomas (median log2 expression 0.00, versus 0.63 in other breast cancers; $p = 0.19$). Activation of a mucin/secretory program is therefore not sufficient to induce REG4; REG4 expression requires a GI lineage context.

Among the remaining cohorts, CHOL showed a gradient consistent with anatomical site: REG4 was higher in distal (median 5.14, $n = 2$) and perihilar (1.94, $n = 4$) tumours than in intrahepatic cholangiocarcinomas (1.10, $n = 30$), although the numbers were too small for formal testing. Histological annotation did not identify mucinous subgroups in LUAD, OV, UCEC or ESCA_EAC, so the relationship between REG4 and mucinous histology could not be tested in these cohorts.

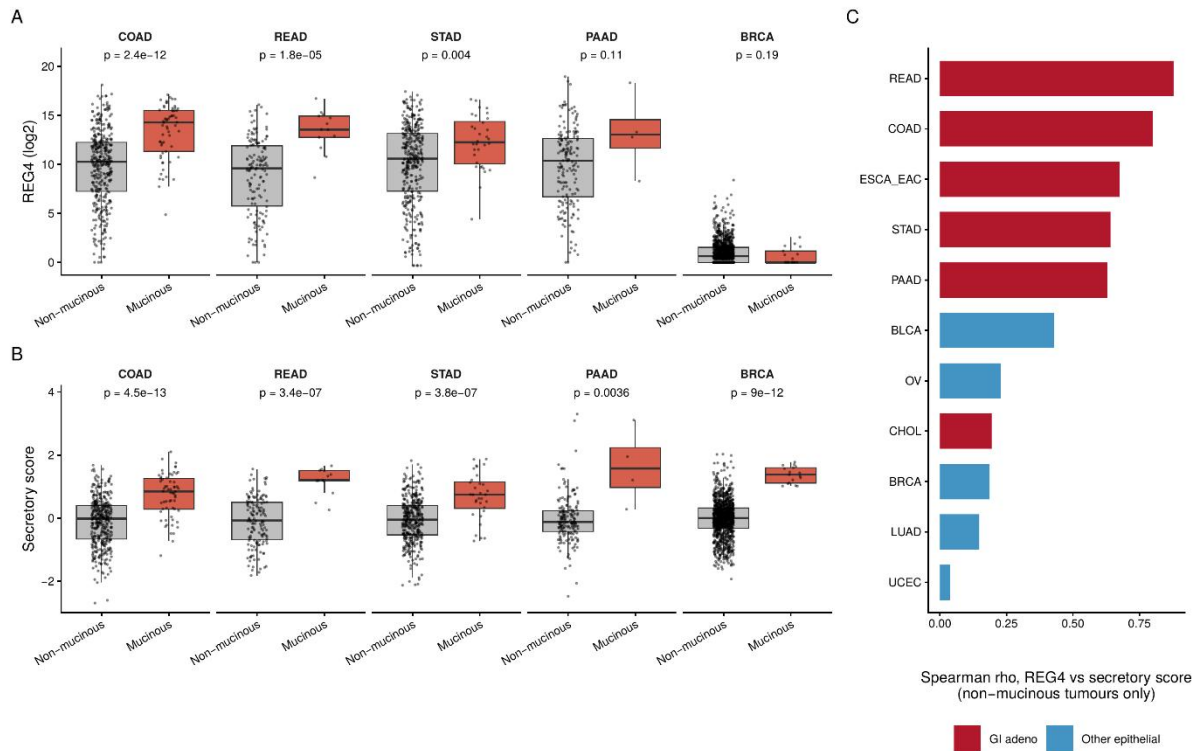


Figure 3. REG4 marks a graded mucinous state that requires GI lineage context. (A) REG4 expression and (B) secretory score in mucinous (including signet-ring and colloid) versus non-mucinous tumours in four GI cohorts and in breast cancer. P values are from Wilcoxon rank-sum tests. Mucinous breast carcinomas activate the secretory program but do not express REG4. (C) Spearman correlation between REG4 and the secretory score after exclusion of all mucinous tumours, shown for GI adenocarcinoma (red) and other epithelial (blue) cohorts.

REG4-associated immune signals in colorectal cancer reflect molecular subtype

Because REG4 and the secretory program were strongly correlated in GI tumours, we asked whether REG4 carries any information about the tumour microenvironment beyond the secretory state it marks. Linear models were fitted for the immune/cytotoxic, NK and stromal/EMT scores, including REG4, the secretory score, their interaction and tumour purity as predictors (Supplementary Table S6).

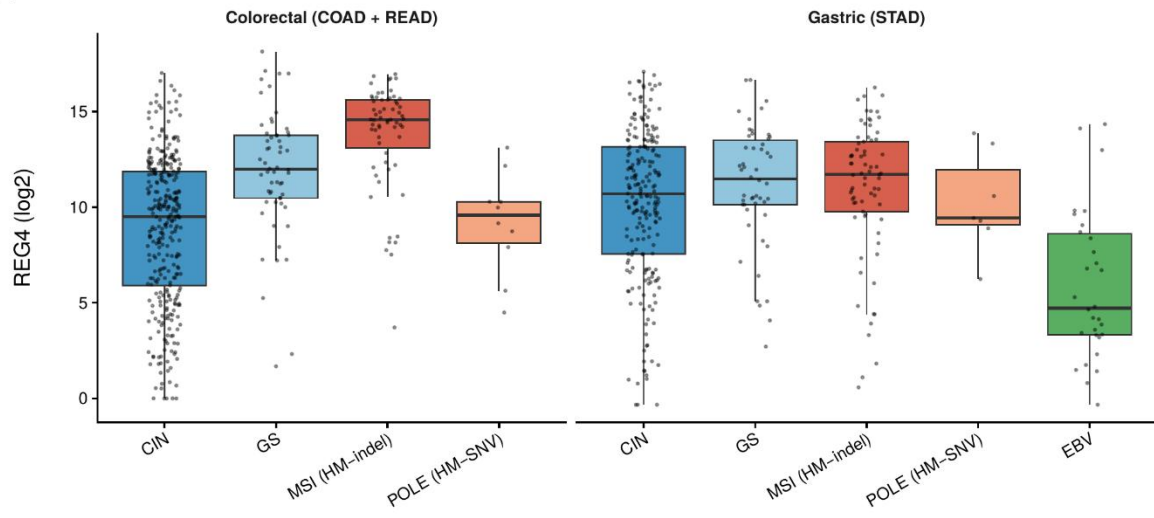
Associations beyond the secretory program were found in only three cohorts. In COAD, REG4 was independently associated with higher immune/cytotoxic ($\beta = 0.28$ per SD, $\text{FDR} = 1.4 \times 10^{-4}$) and NK scores ($\beta = 0.27$, $\text{FDR} = 5.0 \times 10^{-7}$). In READ, REG4 was associated with higher NK scores ($\beta = 0.30$, $\text{FDR} = 0.004$), with a stronger effect in secretory-high tumours (interaction $\text{FDR} = 2.3 \times 10^{-4}$). This was consistent with the NK-specific partial correlations observed earlier in READ ($\rho = 0.38$) and COAD ($\rho = 0.23$). In COAD, tumours that were REG4-high but secretory-low ($n = 39$) had the highest median immune/cytotoxic score of the four REG4/secretory states (0.36, compared with -0.20 in REG4-low/secretory-low tumours). In PAAD, by contrast, REG4 was associated with lower stromal/EMT scores ($\beta = -0.25$, $\text{FDR} = 0.02$), particularly in secretory-high tumours (interaction FDR

$= 1.3 \times 10^{-5}$). REG4 showed no independent microenvironment associations in STAD, ESCA_EAC or PRAD, and only small negative associations in BLCA (β -0.09 to -0.14).

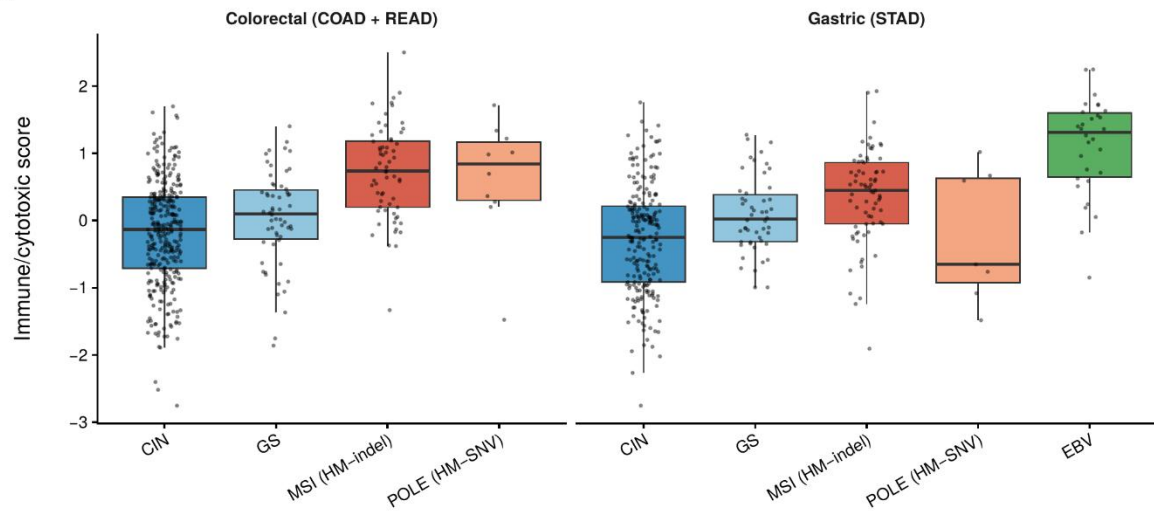
We then asked whether the colorectal immune association reflected molecular subtype, since MSI colorectal cancers are both frequently mucinous and highly immune-infiltrated. REG4 expression differed markedly across PanCanAtlas GI subtypes (Figure 4A). In COAD, REG4 was highest in hypermutated-indel (MSI) tumours (median log2 expression 14.56, $n = 60$), intermediate in genomically stable (GS) tumours (12.07, $n = 49$) and lowest in chromosomally unstable (CIN) tumours (9.50, $n = 224$), an approximately 33-fold difference between MSI and CIN tumours. MSI tumours also had the highest immune/cytotoxic scores (median 0.76, compared with -0.18 in CIN tumours; Figure 4B). Adjustment for molecular subtype reduced the association between REG4 and the immune/cytotoxic score in COAD by approximately 60% (β from 0.28 to 0.11), and the association was no longer significant ($p = 0.096$). The REG4–NK association was attenuated to a similar extent (β from 0.22 to 0.09; $p = 0.055$; Figure 4C). In READ, among tumours with subtype annotation ($n = 117$), REG4 was not significantly associated with either score before or after adjustment. The colorectal immune association of REG4 therefore largely reflects the MSI subtype, in which mucinous differentiation and immune infiltration co-occur, rather than an independent REG4-linked immune program.

Gastric cancer provided a complementary contrast. EBV-positive gastric carcinomas, the most immune-infiltrated gastric subtype (median immune/cytotoxic score 1.31), had the lowest REG4 expression of any GI subtype (median 4.72, $n = 30$), roughly 60-fold lower than CIN (10.69), GS (11.48) and MSI (11.71) gastric tumours (Figure 4A–B). Immune infiltration and REG4 expression are therefore separable: the most immune-rich gastric tumours are REG4-low.

A



B



C

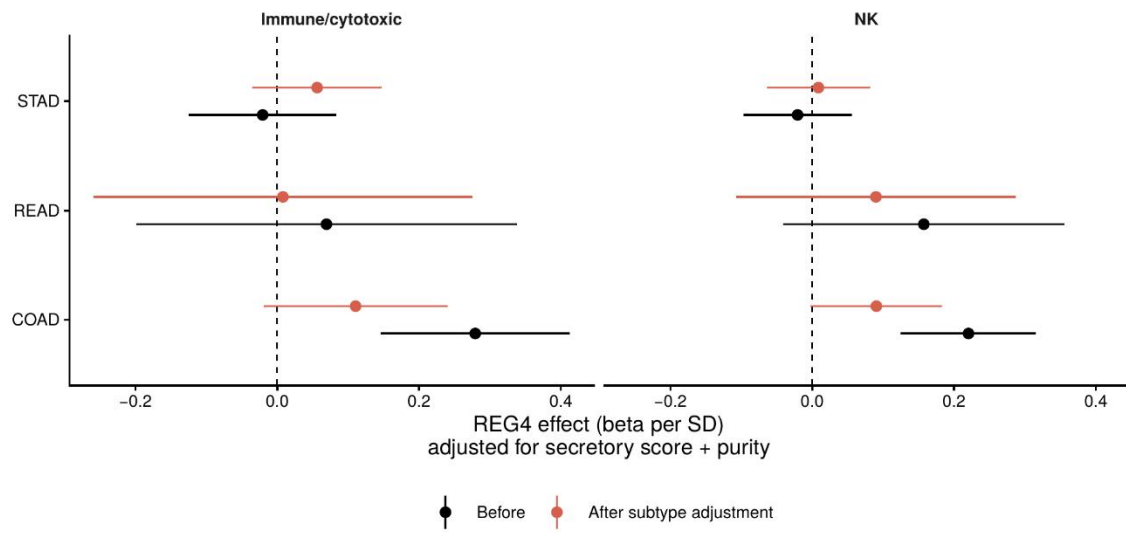


Figure 4. REG4-associated immune signals reflect molecular subtype. (A) REG4 expression and (B) immune/cytotoxic score across PanCanAtlas molecular subtypes in colorectal (COAD and READ combined) and gastric (STAD) adenocarcinomas. CIN, chromosomal instability; GS, genomically stable; MSI (HM-indel), hypermutated-indel/microsatellite-unstable; POLE (HM-SNV), hypermutated-SNV; EBV, Epstein–Barr virus-positive. (C) Effect of REG4 (β per SD, with 95% confidence intervals) on immune/cytotoxic and NK scores in linear models adjusted for secretory score and tumour purity, before (black) and after (red) additional adjustment for molecular subtype.

REG4 is not independently prognostic in GI adenocarcinomas

Finally, we asked whether REG4 carries prognostic information beyond the secretory state it marks. Cox models were fitted in the seven cohorts with epithelial/secretory dominance, comparing a base model (epithelial/secretory score, age and, where available, stage) with the same model plus REG4 (Figure S1; Supplementary Table S5).

In the GI adenocarcinomas, REG4 was not prognostic on its own. Unadjusted hazard ratios for PFI and OS ranged from 0.67 to 1.13 per SD, and none reached significance. After adjustment for secretory state, REG4 did not improve model fit in COAD, READ or ESCA_EAC for either endpoint (likelihood-ratio test $p \geq 0.28$). Two cohorts showed weaker signals. In STAD, REG4 showed a non-significant trend toward worse outcome (PFI HR = 1.20, $p = 0.11$; OS HR = 1.19, $p = 0.095$). In PAAD, REG4 was associated with longer PFI after adjustment (HR = 0.74, $p = 0.04$; likelihood-ratio $p = 0.041$), and the concordance index increased from 0.55 to 0.61. However, this association was not observed for OS (HR = 0.89, $p = 0.47$), and it emerged only after adjustment, while the correlated secretory score shifted toward an adverse effect (HR = 1.77); this pattern is consistent with statistical suppression between correlated predictors. Neither signal remained significant after correction for multiple testing (FDR = 0.33 and 0.27, respectively). In a pooled model across the five GI cohorts, stratified by cohort, REG4 added no prognostic information beyond the secretory score (HR = 0.96 per SD, $p = 0.56$).

BLCA illustrated the difference between a passive marker and an independent predictor. In unadjusted models, higher REG4 was associated with better outcome (OS HR = 0.78, $p = 0.004$; PFI HR = 0.80, $p = 0.011$), but this association disappeared after adjustment for the epithelial/secretory score and stage (OS HR = 0.91, $p = 0.35$; PFI HR = 0.93, $p = 0.44$). The apparent prognostic value of REG4 in BLCA therefore reflects the epithelial state it marks.

PRAD was the only cohort in which REG4 showed an independent association with outcome: higher REG4 was associated with longer PFI after adjustment (HR = 0.71, $p = 0.006$; likelihood-ratio $p = 0.0049$), with the concordance index increasing from 0.58 to 0.62. This association did not reach significance after correction for multiple testing (FDR = 0.064), and the model could not be adjusted for stage or Gleason grade, which are not available in the curated clinical resource for this cohort. The four-state analysis gave consistent results: PRAD tumours that were high for both REG4 and the secretory score had longer PFI than those low for both (HR = 0.42, $p = 0.002$), whereas in the GI cohorts, no four-state contrast that isolated REG4 (REG4-high/secretory-low versus REG4-low/secretory-low) was associated with outcome (Supplementary Table S7).

Taken together, REG4 behaves as a passive marker of secretory state with respect to prognosis in GI adenocarcinomas. Its apparent prognostic associations are explained by the state it marks, with the possible exception of PRAD, which requires validation with full clinicopathological adjustment.

S1

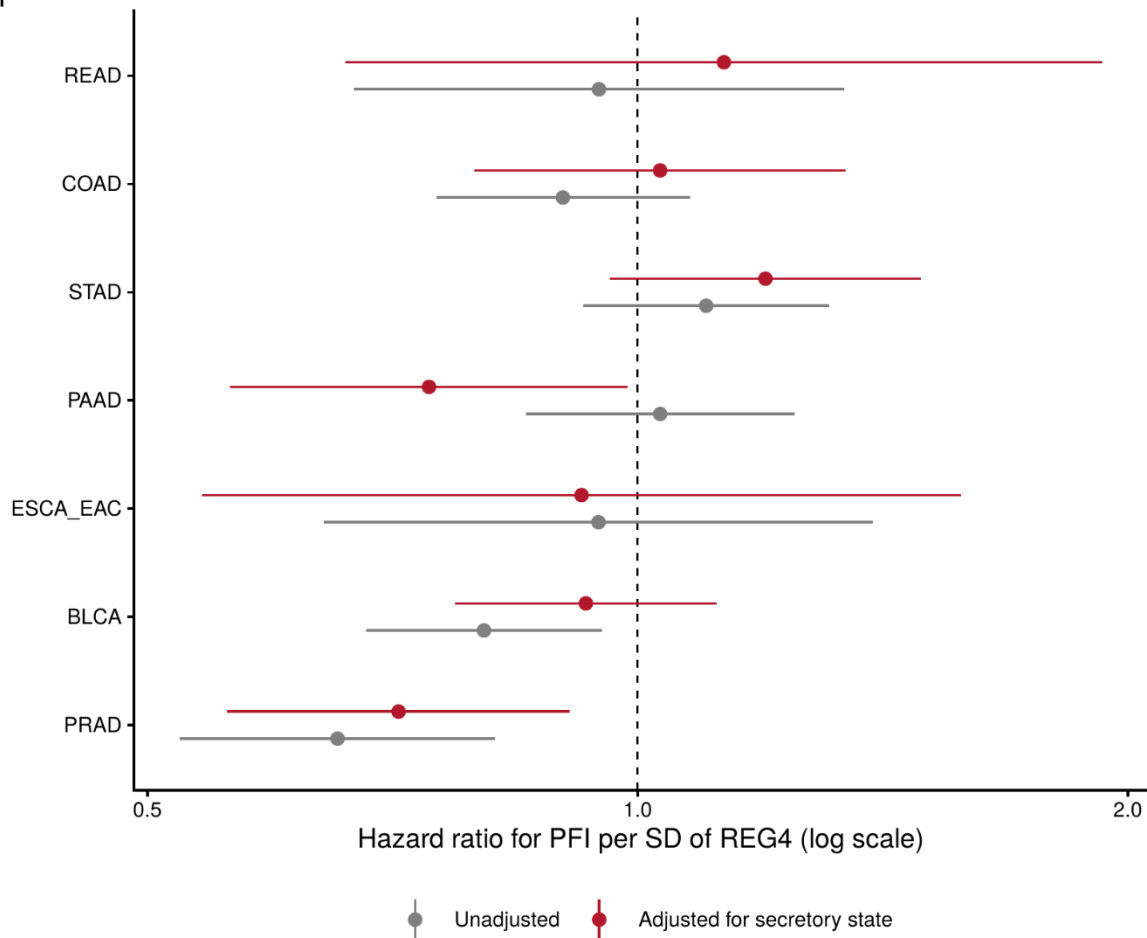


Figure S1. Prognostic value of REG4 in cohorts with epithelial/secretory dominance. Hazard ratios for progression-free interval (PFI) per standard deviation of REG4 expression, with 95% confidence intervals, from unadjusted Cox models (grey) and models adjusted for epithelial/secretory score, age and stage where available (red). The dashed line indicates HR = 1. The x-axis is on a log scale.

Discussion

In this pan-cancer analysis of 9,849 primary tumours, we asked which biological axis REG4 primarily tracks and whether that axis depends on lineage. Four findings emerged. First, REG4 expression is strongly lineage-restricted, being high and nearly universal in GI adenocarcinomas and low or absent in most other tumour types. Second, where REG4 is expressed, its dominant association is with the epithelial/secretory program, and this dominance is specific to GI adenocarcinomas, robust to tumour purity and independent of the precise score definition. Third, REG4 tracks a graded goblet/secretory state that extends beyond mucinous histology, but only in a GI lineage context: mucinous breast carcinomas activate the same secretory program without expressing REG4. Fourth, the immune and prognostic associations of REG4 in GI tumours are largely explained by molecular subtype and secretory state. Together, these results support the view that REG4 is not a pan-cancer immune or stromal marker, but a marker of a lineage-restricted secretory epithelial state.

A notable finding was that the relationship between REG4 and intestinal lineage transcription factors depends on the organ. In colorectal cancer, REG4 was negatively correlated with CDX2, HNF4A and VIL1 but positively correlated with KLF4, a transcription factor required for goblet-cell differentiation. In pancreatic cancer, REG4 was positively correlated with all of these intestinal factors, and in gastric cancer the correlations were weakly positive. We propose that REG4 marks a shift in secretory differentiation relative to each tissue's native state. In the colon, where intestinal identity is native, REG4-high tumours move away from the canonical enterocyte program toward a goblet-like or deep crypt secretory-like state. This interpretation is consistent with the normal role of REG4 as a marker of colonic deep crypt secretory cells and with the known association between reduced CDX2 expression and mucinous, right-sided and MSI colorectal cancers. In the pancreas and stomach, where intestinal identity is not native, REG4-high tumours instead represent the acquisition of intestinal-type differentiation, as seen in intestinal metaplasia of the stomach and in the classical, GI-like subtype of pancreatic ductal adenocarcinoma. Although REG4 has been described as a direct CDX2 target in gastric cancer, its high expression in colorectal tumours with relatively low CDX2 suggests that other secretory-lineage regulators can maintain REG4 expression when canonical intestinal identity is reduced.

The mucinous breast carcinoma result provides a direct test of lineage dependence. These tumours strongly activated the secretory program, including genes such as MUC2 and AGR2, yet did not express REG4. This dissociation shows that mucin production alone is not sufficient to induce REG4, and that REG4 expression requires a GI lineage context that is absent from breast epithelium. This is consistent with earlier immunohistochemical studies reporting that REG4 protein is not detected in breast or lung carcinomas. Conversely, REG4 tracked the secretory program even among non-mucinous GI tumours, indicating that it reports a continuous secretory state rather than a histological category. REG4 may therefore be useful as a quantitative readout of GI secretory differentiation, which is not captured by binary histological classification.

These results also help to explain why REG4 has been linked to diverse and sometimes conflicting tumour features in earlier studies. Two confounders were important in our data. The first is tumour purity: several apparent non-GI associations, including the immune dominance in most squamous cancers and the stromal dominance in ovarian cancer, were lost after purity adjustment. The second is molecular subtype: in colorectal cancer, REG4 was associated with immune and NK signals beyond the secretory program, but approximately 60% of this association was explained by molecular subtype. MSI colorectal cancers were both the highest REG4 expressers and the most immune-infiltrated tumours, providing a common cause for the association. The EBV-positive gastric subtype illustrated the reverse case, being the most immune-infiltrated gastric subtype yet expressing the least REG4. Similarly, REG4 was not independently prognostic in any GI adenocarcinoma, and in bladder cancer an apparent protective association disappeared after adjustment for epithelial state. Studies reporting immune or prognostic roles for REG4 based on unadjusted bulk correlations should therefore be interpreted with caution, as such associations may reflect the cellular state and molecular subtype in which REG4 is expressed.

Several exceptions deserve comment. Prostate cancer was the only non-GI cohort with high and nearly universal REG4 expression, yet REG4 was decoupled from the GI secretory and lineage program there, correlating only modestly with MUC2 and AGR2. REG4 was also the only independent prognostic signal in our analysis, with higher expression associated with longer progression-free interval in primary tumours. This contrasts with earlier reports linking REG4 to hormone-refractory metastatic prostate cancer. The two observations are not necessarily inconsistent, because TCGA primary tumours are largely hormone-naïve, but our prostate result could not be adjusted for Gleason grade or stage and did not survive multiple-testing correction, so it requires validation. Testicular germ cell tumours showed both epithelial/secretory and stromal associations, consistent with the gut-like endodermal elements of non-seminomatous tumours. In pancreatic cancer,

REG4 was associated with lower stromal scores beyond the secretory program, particularly in secretory-high tumours. This may reflect the relationship between the classical epithelial subtype and stromal composition, but it was not tested against published pancreatic subtypes and should be considered exploratory.

This study has limitations. It relies on bulk transcriptomic data from a single resource, TCGA, and therefore cannot establish which cells express REG4 or confirm the findings in independent cohorts. The secretory gene set was defined from known intestinal goblet-cell biology, which may favour colorectal cancer; an unbiased genome-wide co-expression analysis would address this. Program scores were calculated as simple averages of z-scores, and the NK markers overlap partially with cytotoxic T-cell markers. Several cohorts were small (CHOL, n = 36; ESCA_EAC, n = 89), mucinous tumours were few in some cohorts, and histological annotation did not identify mucinous subtypes in lung, ovarian or endometrial cancer. Molecular subtype annotation was unavailable for a substantial fraction of colorectal tumours. Finally, all analyses were at the mRNA level and were observational, so they cannot establish whether REG4 has a functional role in the secretory state it marks.

Future work should address these limitations directly. Single-cell RNA sequencing of GI tumours can determine whether REG4 is expressed by malignant secretory cells, and whether REG4-high malignant cells correspond to goblet-like or deep crypt secretory-like states. Independent cohorts with histological and MSI annotation, together with proteomic data, would test whether the lineage-restricted and subtype-dependent patterns observed here are reproducible at the protein level. Immunohistochemistry of REG4 in mucinous tumours from GI and non-GI sites would directly test the lineage requirement suggested by the breast carcinoma results.

In conclusion, REG4 is not a pan-cancer immune or stromal marker. Its dominant and most coherent association is with a lineage-restricted secretory epithelial state in GI adenocarcinomas, where it tracks goblet-like differentiation in a manner that depends on the organ of origin. Its apparent immune and prognostic associations largely reflect tumour purity, molecular subtype and cellular state rather than an independent REG4-linked program. These findings suggest that REG4 is best interpreted as a quantitative marker of GI secretory state, and they provide a framework for evaluating other state-restricted genes in pan-cancer analyses.